

GANESH GUPTA

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Owned end-to-end design of a multi-agent system using OpenAI Agents SDK and MCP tool-calling: prompt design, reasoning chains, and tool integration with internal systems, powering RAG-based decision support for millions of daily transactions at 99%+ uptime.
- Took the system from architecture through deployed, validated, and observable in production: built a deployment-gated evaluation pipeline for AI governance, enforcing automated compliance, quality, and drift checks, blocking 15% of failed releases.
- Architected AWS and Azure serverless infrastructure (Lambda, S3, DynamoDB, API Gateway, SageMaker) for real-time agentic inference at sub-200ms latency, partnering directly with product stakeholders to ship at scale.

QureAI (AI Startup)

Jan 2024 – May 2025

ML Infrastructure Engineer

- Diagnosed a user-facing RAG retrieval gap directly from real usage signals, then owned the fix end-to-end: redesigned chunking and prompt strategy, lifting recall 61% to 89% and cutting the underlying complaint pattern from 22% to 6% of sessions.
- Led migration of synchronous model serving to async vLLM with autoscaling as an early engineer at this AI startup, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests.

Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

Software Engineer

- Architected and operated 25+ production Java microservices in financial-services distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly financial data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Led standardization of failure-handling contracts across 25+ services in partnership with cross-functional teams through code reviews and design write-ups, reducing production escalations 60% in one quarter.

Projects

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with Pinecone vector retrieval and hybrid re-ranking decomposes multi-table questions into dependency-ordered SQL subqueries; deployable as MCP-compatible server for enterprise data queries.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL

ML/AI Engineering: RAG, Agentic AI, LLM Orchestration, Tool/Function Calling, LangGraph, LangChain, OpenAI Agents SDK, MCP, Vector Databases (Pinecone, FAISS)

Backend & APIs: FastAPI, REST APIs, Node.js, PostgreSQL, Data Pipelines

MLOps & Cloud: AWS (Lambda, EC2, S3, RDS, SQS, SageMaker), Azure, GCP (Cloud Run), Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer